ModelIt!

Student Activity Pack

Designing the Perfect Artificial Organ

Modeling the Bioengineering Challenge of Replacing Living Tissue with Technology

Grade 10L3 | NGSS HS-LS1-6, HS-PS1-3 | Structure and Function

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Why do you think the body sometimes rejects transplanted organs or medical implants?

2 What properties would an ideal material for an artificial organ need to have?

3 Draw what you think happens at the molecular level when the immune system encounters a foreign material inside the body.

4 Why is it so difficult to build artificial versions of organs like the kidney or liver?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Biocompatible Material | Molecular Surface Chemistry | Immune Response Rate | Blood Flow Integration | Structural Durability | Nutrient Transport | Waste Filtering | Electrical Signal Compatibility | Cell Attachment Rate | Functional Lifespan

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Optimal Biocompatibility

Best material with anti-inflammatory surface chemistry

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Immune Rejection Cascade

Less compatible material triggering immune response

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Mechanical Failure

Maximum physiological loading on the implant

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Drug Elution Rate — The controlled release of anti-inflammatory or immunosuppressive drugs from the implant surface to manage local immune response without systemic side effects

Degradation Products — The chemical byproducts released as biodegradable implant materials break down over time — these products must be non-toxic and efficiently cleared by the body

Infection Resistance — The implant's ability to resist bacterial colonization and biofilm formation — infections at implant sites are a leading cause of device failure and can be life-threatening

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why is building an artificial organ that the body does not reject one of the hardest problems in all of engineering — and can we actually do it? Use evidence from your simulation to support your answers.

1. **How does your model demonstrate the fundamental conflict between immune protection and implant acceptance?**

2. **Why is surface chemistry more important than the bulk material for determining immune compatibility?**

3. **What does your model predict about which variable most commonly limits Functional Lifespan?**

4. **How would incorporating living cells into the device change the biocompatibility and functional dynamics?**

5. **What ethical considerations arise when designing artificial organs that some patients can afford and others cannot?**

8 STEM Challenge: Design a Next-Generation Bioartificial Kidney

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YOUR ENGINEERING MISSION

Design a bioartificial kidney concept that combines synthetic filtration membranes with living kidney cells to replace dialysis for end-stage renal disease patients.

THE SCENARIO

Over 800,000 Americans have end-stage kidney disease, with 500,000 on dialysis costing $90 billion per year. Only 25,000 kidney transplants are performed annually due to donor shortages. Your team must design a bioartificial kidney that can be implanted permanently, filtering blood and producing urine without triggering immune rejection — potentially saving millions of lives and billions of dollars.

Engineering Design Thinking

1. WHAT SIZE AND SHAPE WOULD YOUR DEVICE BE, AND WHERE IN THE BODY WOULD IT BE IMPLANTED?
2. HOW WOULD YOU ENSURE THAT LIVING CELLS WITHIN THE DEVICE RECEIVE ADEQUATE NUTRIENTS AND OXYGEN?
3. WHAT IS YOUR STRATEGY FOR PREVENTING BLOOD CLOTTING AT THE DEVICE-BLOOD INTERFACE?
4. WHAT IS THE PROJECTED FUNCTIONAL LIFESPAN AND WHAT DETERMINES WHEN THE DEVICE WOULD NEED REPLACEMENT?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Biomedical Engineers design medical devices, artificial organs, and tissue engineering solutions. They work at medical device companies (Medtronic, Abbott, Boston Scientific), hospitals, research universities, and startups, earning $70,000-$150,000/year. Tissue Engineers who grow living replacements for damaged organs represent one of the most innovative frontiers in medicine, earning $80,000-$160,000/year.